MIL-P-48357A (AR)
26 October 1977
SUPERSEDING
MIL-P-48357
7 June 1974

MILITARY SPECIFICATION

PROPELLANT, DOUBLE-BASE SHEET, M36

This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a Double-Base Sheet Propellant, M36 referred to herein as the "propellant".

1.2 Classification

Type I 0.020-0.040 inch thick
Type II 0.095-0.115 inch thick

2. APPLICABLE DOCUMENTS

2.1 Issue of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

JAN-W-181 - Wax (Candelilla)
MIL-N-244 - Nitrocellulose (For Use in Explosives)
MIL-N-246 - Nitroglycerin
MIL-N-3399 - 2-Nitrodiphenylamine
MIL-D-22346- Di-n-propyl adipate
MIL-A-48078- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-B-48356- LC-12-6 Ballistic Modifier

FSC: 1376

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants: Sampling, Inspection and Testing
MIL-STD-652 - Propellants, Solid for Cannons Requirements and Packing
MIL-STD-1171- Acceptance and Description Sheets for Propellants and Explosives

DRAWINGS

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 CFR Parts 100-199 Department of Transportation Rules and Regulations for Transportation of Explosives and Other Dangerous Articles

The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite:

49 CFR Parts 100-199 (latest revision)

3. REQUIREMENTS

3.1 Preproduction lot. This specification makes provisions for preproduction inspections and testing. Submission of preproduction quantity by the contractor shall be as specified in the contract (see 4.2).

3.2 General Requirement. Unless otherwise specified, the propellant shall be in accordance with the requirements specified herein and the applicable documents listed in Section 2.

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3.2.1 Materials. The component material used in the manufacture of the propellant shall comply with the following requirements.

Component Material	Conforming to
Nitrocellulose (12.6% Nitrogen)	MIL-N-244, Grade A, Type I (Processed from Cotton Linters Cellulose)
Nitroglycerin	MIL-N-246, Type I
Di-n-propyl adipate	MIL-D-22346
Wax (Candelilla)	JAN-W-181
2-Nitrodiphenylamine	MIL-N-3399
Ballistic Modifier (LC-12-6)	MIL-B-48356

3.2.2 Dimensions. The dimensions of the propellant sheet shall be as follows:

(a) Thickness of component sheet:

Type I	0.020-0.040 inch
Type II	0.095-0.115 inch

(b) Width of sheet (perpendicular to direction of rolling) shall be as specified in the contract or purchase order.

(c) Length of sheet (parallel to direction of rolling) shall be as specified in the contract or purchase order. (Thickness, width and length may vary within specified limits in any single sheet).

3.3 Performance and Product Characteristics. The propellant shall meet the following performance and product characteristics requirements:

3.3.1 Composition. The analytical composition of the propellant shall be determined on a moisture free basis and shall be as specified below:

Component Material	Percent by Weight
Nitrocellulose (12.6% nitrogen)	49.0 ± 1.5
Nitroglycerin	40.6*
Di-n-propyl adipate	3.0*
2-Nitrodiphenylamine	2.0 ± 0.5
LC-12-6 Ballistic Modifier	5.3*
Candelilla Wax	0.1*

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*Percent indicated is nominal. Manufacturer may vary values to the extent required to produce propellant to meet requirements of Section 3.

3.3.2 Moisture content. The moisture content of the propellant shall be not more than 0.6 percent.

3.3.3 Heat of explosion. The mean heat of explosion and standard deviation of the propellant shall resolve to a point which, when plotted, lies within the region bounded by the acceptance curve (see figure 2).

3.3.4 Strand burning rate. Inhibited strands of the propellant shall meet the following strand burning rate requirements:

3.3.4.1 Average burning rate. The average burning rate of the propellant should be within the desired specification limits of Table I, when the burning is conducted under the controlled conditions, of pressure and temperature as specified in Table I. If the average burning rates exceed the upper limit or are exceeded by the lower limit, the sum of the squared differences between the actual and specified values shall not be greater than 0.00085 $(in/sec)^2$.

TABLE I

	Burning Rate (in/sec)			
	-40°C (-40°F)		63°C (145°F)	
Pressure,psia	Lower Limit	Upper Limit	Lower Limit	Upper Limit
1600	0.84	1.07	1.10	1.41
2000	0.89	1.14	1.18	1.42
2400	0.93	1.18	1.20	1.45
2800	0.95	1.21	1.18	1.43
3200	0.92	1.21	1.10	1.39
3600	0.78	1.17	1.10	1.37

3.3.4.2 Average burning rate range. The average burning rate range (the arithmetical average of the difference between the highest and the lowest burning rate obtained at each pressure temperature condition) shall not exceed 0.060 inch per second.

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3.3.5 Taliani stability. The propellant shall be stable when placed in an inert atmosphere at a temperature of 110 degrees C. Evolved gas pressure shall not exceed a rate of 1.1 millimeters of mercury per minute when judged in accordance with the slope of the pressure time curve at 100 minutes as determined by Test Method 406.1 (T) in Standard MIL-STD-286. The sample shall be taken in accordance with 4.5.4.

3.3.6 Workmanship. The propellant sheet, including all material used therein shall be manufactured using the best commercial practices and finished in a manner to assure compliance with this specification and other related documents herein and shall be homogeneous and free of all contamination, including avoidance to exposure to direct sunlight and acid fumes.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 60 samples of sheet propellant in accordance with instructions issued by the Contracting Officer.

4.3.2 Inspections to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.4 and 4.5 of this specification and any or all requirements of the applicable drawings, and/or documents.

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4.3.3 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with lot formation provisions of MIL-A-48078. In addition, inspection lots of Propellant, Double-Base, Sheet shall contain:

a. Double-Base Sheet Propellant from not more than one lot from one manufacturer.

4.4.2 Examination. Unless otherwise specified in the Classification of Defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II (See MIL-A-48078).

4.4.3 Testing

4.4.3.1 Materials and Performance: Each lot of propellant shall be sampled and tested for composition, form, dimensions and chemical-physical properties in accordance with this specification and other documents as specified in Table II below:

TABLE II

Property or Material	Type of Defect	Code Number	Requirements Paragraph	Test Paragraph
Nitrocellulose	Major	01001	3.3.1	4.5.2
Nitroglycerin	For Information	01002	3.3.1	4.5.2.1
2-Nitrodipheny-lamine	Major	01003	3.3.1	4.5.1
Di-N-propyl-adipate	For Information	01004	3.3.1	4.5.2.2
LC-12-6 Bal Mod	For Information	01005	3.3.1	4.5.2.3
Candelilla Wax	For Information	01007	3.3.1	4.5.2.4

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Moisture	Major	01006	3.3.2	4.5.3
Heat of Explosion	Major	01008	3.3.3	4.5.5
Strand Burning Rate	Major	01009	3.3.4	4.5.6
Average Burning Rate Range	Major	01010	3.3.4.2	4.5.6.2.3
Taliani Stability	Major	01011	3.3.4	4.5.4

4.4.4 Inspection equipment. The government reserves the right to inspect the contractor's equipment and determine that he has available and utilizes correctly, measuring and test equipment of the required accuracy and precision and that the instruments are of the proper type and range to make measurements of the required accuracy. Commercial inspection equipment, shall be employed where applicable for all tests and examinations specified in 4.4 and 4.5. The contractor is responsible for assuring proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purpose (see 6.3).

4.4.5 Sampling

4.4.5.1 Sampling for acceptance tests. Unless otherwise specified the sampling plans and procedures used in determining the acceptability of the propellant shall be in accordance with Standard MIL-STD-105. Sixty samples shall be selected for acceptance testing. The samples shall be eleven inches long by four inches wide and shall be cut from each of 60 individual even speed rolled sheets selected at random from each lot of propellant submitted for acceptance. The long axis of each sample shall lie in a direction which is parallel to the final (even-speed) rolling. The samples shall be numbered from 1 to 60 to facilitate testing.

4.4.5.2 Division of samples for laboratory tests. From each sample of 4.4.5.1, four types of test specimens shall be prepared in accordance with Figure 1 and shall be designated as follows:

Test Sample A — For moisture determination (4.4.5.1 and Figure 1)

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Test Sample B	For heat of explosion determination. (The dimensions of these specimens will be approximately 11 inches long by 1.5 inches wide.) (4.4.5.1 and Figure 1)
Test Sample C	For strand burning rate determination (4.4.5.1 and Figure 1)
Test Sample D	For chemical analysis and stability test (4.4.5.1 and Standard MIL-STD-286).

4.4.6 Acceptance tests. Test samples selected and prepared in accordance with 4.4.5 shall be subjected to the tests in 4.5. Failure of any sample to comply with any of the requirements specified herein shall be cause for the rejection of the lot represented by the sample.

4.4.6.1 Composition test. A chemical composition analysis shall be performed on each lot submitted for Government acceptance. The chemical analysis shall be performed at a laboratory designated, in the contract or purchase order, by the procuring activity. (see 6.2)

4.5 Test Methods and Procedure. The test methods and procedures shall be used for determining conformance of the propellant with the requirements of 3.3.

4.5.1 2-Nitrodiphenylamine. The percentage of 2-Nitrodiphenylamine in sample shall be determined by Method 218.4 of MIL-STD-286 except that a diluted aliquot of the solution prepared for the determination of Di-n-propyl adipate by the infrared method shall be used (see 4.5.2.2)

4.5.2 Nitrocellulose Determination. The Nitrocellulose content shall be determined using MIL-STD-286, Method 209.2. The calculation for the percent of Nitrocellulose is as follows:

% Nitrocellulose = 100 - (% Nitroglycerin + % Di-n-propyl adipate + % 2-nitrodiphenylamine + % LC-12-6 + % Candelilla Wax)

4.5.2.1 Nitroglycerin. The percentage of Nitroglycerin shall be determined in accordance with MIL-STD-286, Method 208.1.

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4.5.2.2 Di-n-propyl-adipate. The percentage of Di-n-propyl adipate shall be determined in accordance with MIL-STD-286, Method 222.1.

4.5.2.3 LC-12-6 Ballistic Modifier.-The percentage of LC-12-6 shall be determined in accordance with MIL-STD-286, T-316.1 (Atomic Absorption to determine lead content). The calculation for the percent of LC-12-6 is as follows:

$$\% \text{ LC-12-6} = \frac{\% \text{ Lead found} \times 100}{\text{wt. \% lead in LC-12-6}}$$

4.5.2.4 Candelilla Wax. The percentage of wax shall be determined using the following method and procedure and using the equipment and material as outlined below.

4.5.2.4.1 Sample preparation. The test sample shall be milled to pass a 20 mesh screen. A 10 gram sample shall be accurately weighed. The sample shall be added to a Soxhlet

[redacted]

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4.3 Quality Conformance I

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4.5.3.1 Sample Preparation. The cutting, weighing and transferring to a moisture tight container, must be done rapidly to minimize the change in moisture content through exposure to the atmosphere (see figure 1).

4.5.3.2 Acceptance criteria. The lot shall be rejected if the propellant sample fails to meet the requirements of 3.3.2 when subjected to the test of 4.5.3.

4.5.4 Taliani stability test. A section weighing a minimum of 4.0 grams shall be taken from each of the 60 chemical analysis and stability test samples (Test Sample D of 4.4.5.2). The samples shall be reduced in size in a Wiley Mill, or equivalent, and blended thoroughly to give a uniform composite sample weighing at least 150 grams. The composite sample shall be stored in a moisture-tight container and properly identified; for example:

Test Sample D (See Figure 1)

No. 1

M36 Propellant Lot #

4.5.4.1 Test method. The stability test shall be performed using the apparatus and procedure specified in Method 406.1 of Standard MIL-STD-286.

4.5.4.2 Acceptance criteria. The lot represented shall be rejected if the propellant sample fails to meet the requirements of 3.3.5 when subjected to the test of 4.5.4.1.

4.5.5 Heat of Explosion

4.5.5.1 Sample Preparation. Ten Heat of explosion test samples (Test Sample B of 4.4.5.2) shall be selected at random from the 60 test samples of 4.4.5.1 for heat of explosion tests. From each of the samples selected, a portion weighing at least 20 grams shall be reduced in size (by cutter or grinder) to pass a U.S. No. 20 mesh screen. Each of these portions shall be stored in a moisture tight container and properly identified.

4.5.5.2 Test Method. The heat of explosion test shall be performed using the apparatus and procedure as specified in Method 802.1 of MIL-STD-286. Two heat of explosion determinations shall be made on each sample of propellant prepared.

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4.5.5.3 Acceptance criteria. Failure of the propellant sample to meet the requirement of 3.3.3 shall be cause for the rejection of the lot. The average heat of explosion and the standard deviation shall be determined from the duplicated results obtained from each sample (see 6.5).

4.5.6 Strand Burning Rate. The 60 strand burning rate test samples (Test Sample C of 4.4.5.2) shall each be cut in strands of 1/8 to 3/16 inch wide by 7 1/2 inches in length. The strands shall be stored in moisture tight containers and properly identified.

4.5.6.1 Test Method. The 60 strands prepared in accordance with 4.5.6 shall be tested as specified in MIL-STD-286, Method T803.1. The strand burning rate test shall consist of burning five inhibited strands under each condition of pressure and temperature shown in Table I (3.3.4.1). The clock readings for the burning rates shall be recorded to the nearest 0.005 second.

4.5.6.2 Evaluation of test results

4.5.6.2.1 Individual strands. If the strands are burned in increments, the strands shall be divided into equal increments (three preferred) and the burning rate of each of the increments of each strand shall be determined in inches per second. The value obtained shall be considered acceptable for burning rate computations if each of the following conditions are met:

(a) The difference between the burning rates of the fastest and slowest burning increments of the three equal length increments of any strand is not more than 0.056 inches per second.

(b) The difference between the burning rates of the two increments that agree most closely is not more than 0.047 inch per second.

NOTE: If both values are exceeded, a second strand may be burned to replace faulty data due to test equipment.

4.5.6.2.2 Groups of strands. Groups of strands burned at same temperature-pressure condition, regardless of the number of increments per strand, the range of burning rates for each

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group of five complete strands burned at the same temperature-pressure condition shall be evaluated by dividing the range of the four closest burning rates by the range of all five burning rates. The values obtained shall be considered acceptable for computation of range of burning rate, if the ratio so obtained is less than 0.358.

NOTE: If this value is exceeded, an additional strand may be burned to replace faulty data due to test equipment.

4.5.6.2.3 Calculation for average burning rate. The average burning rate for each of the twelve conditions of temperature pressure shall be determined from the five acceptable burning rate results obtained at the same temperature pressure conditions. The average burning rate so obtained shall be compared with the respective burning rates of Table I (3.3.4.1).

4.5.6.2.4 Burning rate range. The range of individual values for strand burning at each of the twelve conditions of burning shall be determined by subtracting the lowest rate obtained in each group of five strands from the highest rate obtained in the respective group. The arithmetic average range of all twelve groups shall be computed.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking

5.1.1 Packing. Level C. Packing shall be in accordance with Code of Federal Regulations CFR 49, Parts 100-199."

5.1.2 Marking. Level C. Marking shall be in accordance with the Code of Federal Regulations and Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The rocket propellant covered by this specification is a high energy double-base sheet propellant and is intended for use in the manufacture of propellant for the Dragon program.

6.2 Ordering data. Procurement documents shall specify the following:

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a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Dimensions (see 3.2.2)
d. Process. Details of the manufacturing process and the equipment used by the Contractor will be submitted to the procuring activity in writing prior to commencement of manufacture.
e. Acceptance and description sheets shall be prepared for each lot in accordance with MIL-STD-1171.

6.3 Submission of test data. In addition to the normal distribution of records, one copy of all test data shall be forwarded to: Commander, ARRADCOM, ATTN: DRDAR-QAR-R, and one copy to: ATTN: DRDAR-LCU-EP, Dover, New Jersey 07801.

6.4 Inspection code number. The five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.5 Surveillance sample. Samples shall be forwarded to ARRADCOM, Dover, New Jersey, ATTN: DRDAR-LCE, for the propellant surveillance test program in accordance with MIL-STD-652.

Custodian:
ARMY-AR

Preparing Activity:
ARMY-AR

Project Number: 1376-A089

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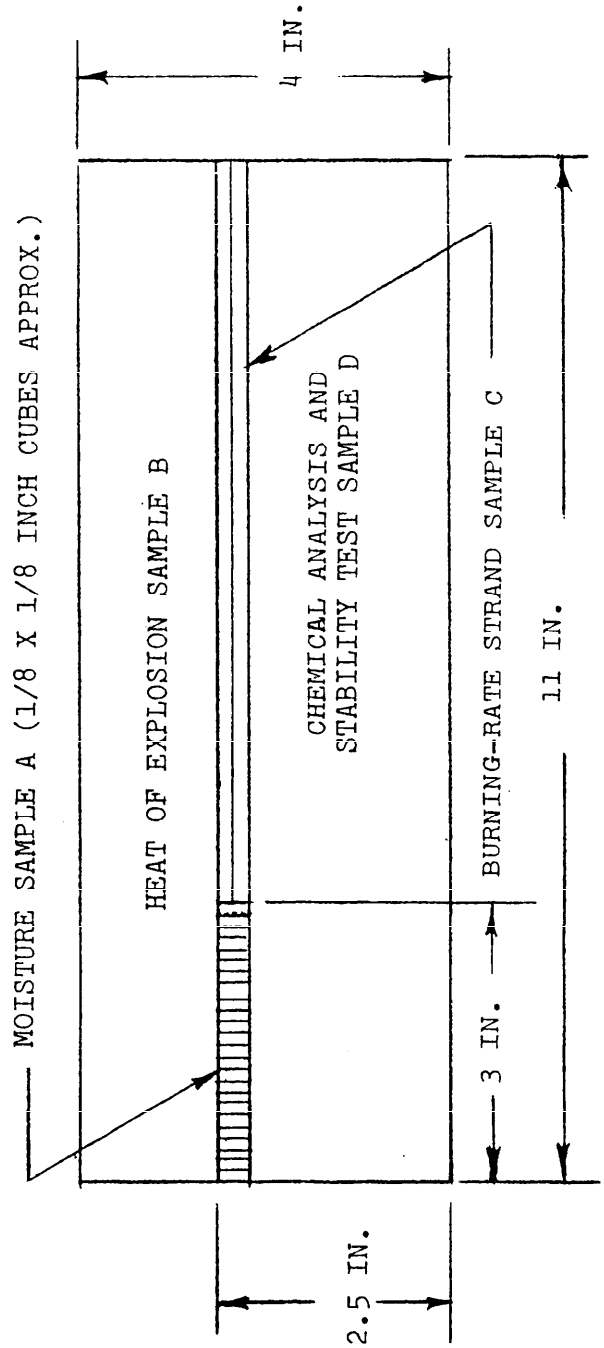


FIGURE 1. DIVISION OF SAMPLE SECTIONS FOR LABORATORY TESTS

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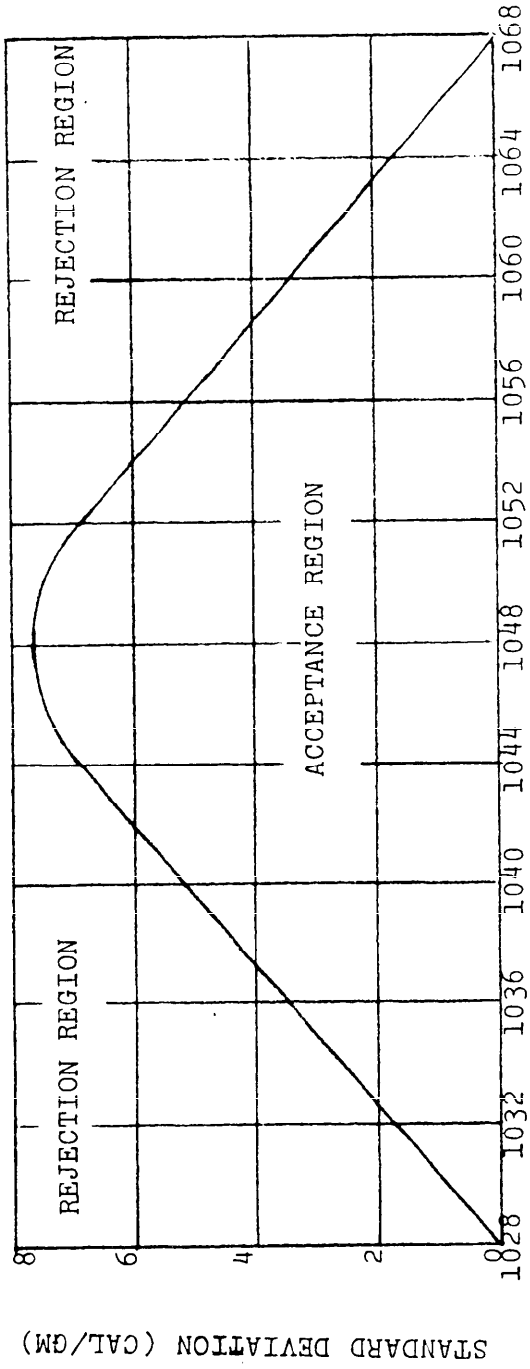


FIGURE 2. ACCEPTANCE CHART FOR HEAT OF EXPLOSION REQUIREMENT

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING.

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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